

- ① Translation of a statement into compiler phases.
- ② Language processing system & compiler phases [Theory]
- ③ Error finding types of error:

Three types of Runtime error

"Special logical error"

└─ if (x > 10)
 printf("Greater than 20");
└─ logical language error

④ Lexeme | Pattern | Token

⑤ Predictive parsing (Left Recursion, Left Factoring,
First & Follow, Predictive Parsing
Table, LL(1) Grammar)

⑥ NFA to DFA using Subset Construction

⑦ Recursive Descent Parsing

⑧ Ambiguous Grammar

Recursive Descent Parsing

Input: aabbde

Grammar:

$S \rightarrow Aa \mid Ce$

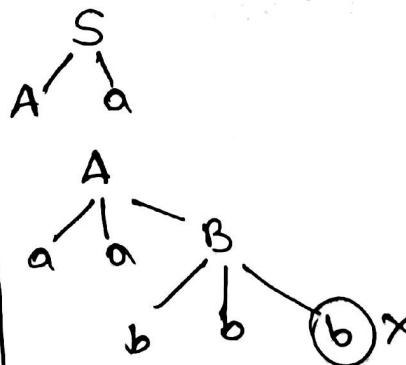
$A \rightarrow aab \mid aaba$

$B \rightarrow bbb$

$C \rightarrow aaD$

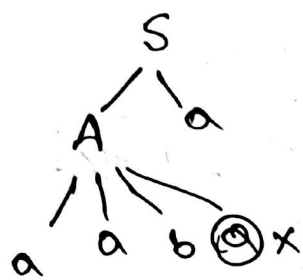
$D \rightarrow \text{~~bbb~~} bbd$

STEP 1 a a b b d e



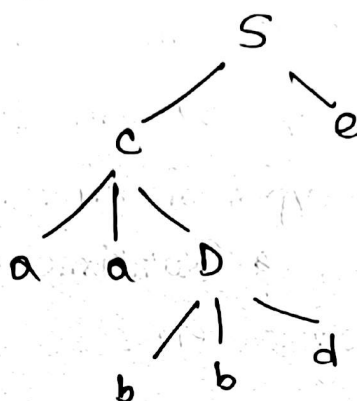
Failure

STEP-2: aabbde



Failure

STEP-3 aabbde



Success

$P \rightarrow i | c | nTS$
 $Q \rightarrow P | aS | bQc$
 $R \rightarrow b | \epsilon$
 $S \rightarrow c | Rn | \epsilon$
 $T \rightarrow Sq | R$

Non-Terminal	First	Follow
P	i, c, n	\$, c
Q	i, c, n, a, b	c
R	b, ϵ	n
S	c, b, n, ϵ	\$, c, q
T	c, b, n, q, ϵ	c, b, n, \$

Parsing Table

	i	c	n	a	b	q	\$
P	$P \rightarrow i$	$P \rightarrow c$	$P \rightarrow nTS$				
Q	$Q \rightarrow P$	$Q \rightarrow P$	$Q \rightarrow P$	$Q \rightarrow aS$	$Q \rightarrow bQc$		
R		$R \rightarrow \epsilon$	$R \rightarrow \epsilon$	.	$R \rightarrow b$ $R \rightarrow \epsilon$		$R \rightarrow \epsilon$
S		$S \rightarrow c$ $S \rightarrow \epsilon$	$S \rightarrow Rn$		$S \rightarrow Rn$	$S \rightarrow \epsilon$	$S \rightarrow \epsilon$
T		$T \rightarrow Sq$ $T \rightarrow R$	$T \rightarrow Sq$ $T \rightarrow R$		$T \rightarrow Sq$ $T \rightarrow R$	$T \rightarrow Sq$	

ghaban | ③

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow ID$$

~~$$ID * ID * ID$$~~

$$ID + ID * ID$$

$$T \rightarrow ID$$

$$E \rightarrow E + T$$

$$E \rightarrow E * T$$

$$E \rightarrow T$$

$$T \rightarrow ID$$

Demo Question $\rightarrow$ Assignment